Creating Cohorts of Songs Using Machine Learning

Project Report

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1. Executive Summary

This project aims to create cohorts (clusters) of songs using Spotify song features. The objective is to discover groups of similar songs that can improve recommendation systems.

2. Problem Statement

Spotify seeks to improve user engagement by recommending relevant songs. This project uses exploratory data analysis, feature engineering, dimensionality reduction, and clustering techniques to identify similar song groups.

3. Dataset Description

The dataset contains Rolling Stones songs available on Spotify along with musical features such as acousticness, danceability, energy, loudness, tempo, valence, popularity, and duration.

4. Data Cleaning

- Missing value analysis
- Duplicate removal
- Outlier detection
- Data type corrections

5. Feature Engineering

- Release Year extraction
- Duration in minutes
- Popular song indicator
- Additional derived features

6. Exploratory Data Analysis

- Album-wise popularity analysis
- Popularity distribution
- Correlation analysis
- Energy vs Popularity
- Danceability vs Popularity
- Popularity trend over years

7. Dimensionality Reduction (Principal Component Analysis - PCA)

The dataset contains several numerical features describing different musical characteristics of songs, such as danceability, energy, acousticness, loudness, tempo, and valence. Many of these features may be correlated with one another and may contain redundant information. Using all features directly for clustering can increase computational complexity and may affect the quality of the generated clusters.

To address this issue, Principal Component Analysis (PCA) was applied as a dimensionality reduction technique. PCA transforms the original features into a smaller set of uncorrelated variables called principal components while retaining most of the information present in the dataset. This helps reduce noise, eliminate redundancy, and improve clustering efficiency. Additionally, PCA facilitates better visualization of the data and enables clustering algorithms to identify more meaningful patterns among songs.

8. Cluster Analysis

The primary objective of cluster analysis is to group songs with similar musical characteristics into distinct cohorts. Since no predefined labels are available, clustering is performed using unsupervised machine learning techniques. The resulting clusters represent groups of songs that share similar audio features and listening characteristics.

A. Identification of the Optimal Number of Clusters

Before applying any clustering algorithm, it is necessary to determine the appropriate number of clusters. The Elbow Method was used to analyze the Within-Cluster Sum of Squares (WCSS) for different values of K. The point where the reduction in WCSS begins to slow significantly indicates the optimal number of clusters. In addition, the Silhouette Score was calculated to evaluate cluster quality by measuring the separation and compactness of clusters. Based on these methods, the optimal number of clusters was selected for further analysis.

B. Application of Clustering Algorithms

After determining the optimal number of clusters, clustering algorithms were applied to segment the songs into meaningful groups. K-Means Clustering was used as the primary clustering technique due to its efficiency and ability to create compact clusters. Hierarchical Clustering was also performed to analyze the hierarchical relationships among songs and to validate the clustering structure. These algorithms grouped songs according to similarities in their musical attributes.

C. Model Comparison and Selection

The performance of the clustering algorithms was evaluated using metrics such as the Silhouette Score. Higher Silhouette Scores indicate better cluster separation and cohesion. By comparing the results of K-Means and Hierarchical Clustering, the algorithm that produced the most meaningful and well-defined clusters was selected as the final model. This comparison ensured that the clustering solution was both statistically sound and practically useful.

9. Cluster Profiling

After generating the clusters, each cluster was analyzed to understand its unique characteristics. The average values of important features such as energy, danceability, acousticness, instrumentalness, valence, and popularity were calculated for each cluster. Based on these feature patterns, meaningful business labels were assigned to the clusters.

For example, clusters with high energy and danceability were identified as Party Songs, while clusters with high acousticness and low energy were categorized as Acoustic Songs. Similarly, clusters with high instrumentalness were labeled Instrumental Songs, and clusters with high valence were interpreted as Happy Songs. Cluster profiling transforms technical clustering results into meaningful business insights that can be easily understood and utilized.

10. Business Recommendations

The identified song cohorts provide valuable opportunities for enhancing Spotify's recommendation system. Songs belonging to the same cluster can be recommended to users who have shown interest in similar songs, thereby improving recommendation relevance and user satisfaction.

The clusters can also be used to create mood-based playlists such as Party, Acoustic, Instrumental, and Happy playlists. These personalized playlists can enhance user engagement and listening experience. Furthermore, understanding song cohorts allows Spotify to better analyze user preferences, improve content discovery, and increase overall platform retention through more targeted recommendations.

11. Conclusion

This project successfully applied exploratory data analysis, feature engineering, dimensionality reduction, and clustering techniques to identify meaningful cohorts of songs. PCA helped reduce the dimensionality of the dataset while preserving important information, thereby improving clustering performance. Various clustering algorithms were evaluated, and the most suitable model was selected based on cluster quality metrics.

The resulting clusters revealed distinct categories of songs characterized by their musical attributes and popularity patterns. These findings demonstrate how machine learning techniques can be effectively used to support recommendation systems and personalized music experiences. The developed song cohorts can serve as a foundation for advanced recommendation engines, playlist generation systems, and future music analytics applications.